treatment of AIDS could be out by 2003. One of the daunting challenges faced by researchers in fighting the HIV virus has been the inability of drugs to attach themselves to the virus and stop it from reproducing any further. Nanotechnology-powered medical techniques like buckyballs have the ability to fix themselves to the virus, thus preventing further reproduction.

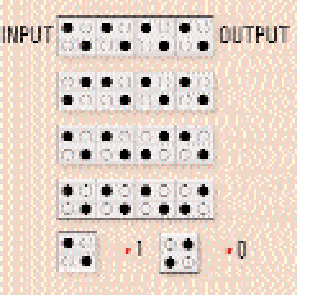
Yet another exciting (though futuristic) prospect that Nanotechnology presents is the ability to have minute machines travelling inside our body protecting us from the inside. These 'nanodoctors' will be able to find and repair damage at the cellular level. For this to be possible, molecular assemblers—with better capabilities than the current scanning tunnelling microscopes—are needed. They have to have the capability to move atoms and molecules faster and more precisely than present day STMs. However, these nanodoctors won't be calling on you for the next 15-20 years, to say the least.

The concept of Nanotechnology-powered doctors inside the human body has a long way to go before it can become a reality. The Centre of Biotechnology at the University of Michigan is working on a technology that could diagnose and monitor malignant cancer cells. This is more of a monitoring use of Nanotechnology for detecting and monitoring tumours. (*See illustration, 'Monitoring of Malignant Cells' for more details.*) The technology is aimed to target cancer cells and sometimes even suggest cures. It will then monitor the effect of the medication taken by the patient on the cells.

One such attempt is being made by IBM's Dr Christoph Gerber, one of the inventors of AFM on biological cells. His group makes use of an AFM's imaging tip along with ultra-small lasers. The tips are attached to cantilevers, which are small springs with movements that help measure the force of each atom. Another researcher, Peter Vettiger, from IBM's Zurich Research Laboratory, believes that this movement can be used to build nanomachines that will be able to act independently. These nanomachines could be used to deliver treatment to cells in the form of microcapsules. For example, specific proteins of cancer cells will be producing a certain amount of force which will help identify that cell. The nanomachine will detect that cell and attach itself to the cell. Then it can release the right amount of anti-cancer drug to the exact location.

Quantum Dot Cell Operation

Quantum dot cells are basically small semiconductor or metal boxes that hold a defined number of electrons in them. Remember that small means cells just a few nanometers across. Electrical fields or insulation is used to hold electrons in place in these dot cells. A string of dot cells one next to the other is proposed to act as a wire across which data can be sent.



As you can see from the figure, each cell has two electrons in them with two of the quantum dot cells left unused. This arrangement of quantum dot cells was proposed by Lent and Porod of Notre Dame. Now if a signal is applied to the left end, the signal propagates itself through the wire as shown. It simply passes on through each of the cells when one cell exerts a force on its neighbouring one to pass on the signal.

Lent and Porod have also come up with an inverter gate which will produce the opposite of the input signal as the output. Combining the inverters and other forms of gates will eventually make nanocomputers a reality soon.

Life after death?

Did you dismiss Austin Powers coming back to life after 30 frozen years as science fiction that happens only in movies? Welcome back to reality. People are already queuing up to have their bodies frozen in cryonic suspension facilities after they die, hoping they'd be back to see life again in the future. Cryonic suspension basically freezes dead bodies to –320^{o} Fahrenheit using liquid Nitrogen. The body fluids are replaced by a glycerine-based solution, which protects the body systems from damage due to freezing. However, remember that Nanotechnology won't be bringing you back to life anytime soon.

Many of the concepts that Nanotechnology presents may look impossible now, but they may not be so far away after all. Could any of you who've been through the 1970s have predicted that the computers of 2001 would have been so powerful? Or would you even have thought of something like a television in the 1960s? Likewise, Nanotechnology is nearer than you can think. The nano-storm will catch you quietly. The only difference being that it will come in a silent and subdued manner, much like how we embraced and used artificial fibres over the years without knowing it. And it will make a tremendous impact on our lives.

http://www

www.foresight.org/
Foresight Institute answers all the questions you might have about Nanotechnology

www.techreview.com/nanotech.asp
Look up Techreview's Nanotechnology site for the latest in Nanotechnology

www.foresight.org/EOC/index.html
Have a look at the book which started it all—Eric Drexler's *Engines of Creation*.